

Mark schemes

Q1.

(a) (i) $\overrightarrow{AB} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad \overrightarrow{AD} = \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$
either $\overrightarrow{AB}$ or $\overrightarrow{AD}$

B1

$$\begin{vmatrix} \mathbf{i} & 2 & 4 \\ \mathbf{j} & -1 & 3 \\ \mathbf{k} & 3 & -1 \end{vmatrix} = \begin{pmatrix} -8 \\ 14 \\ 10 \end{pmatrix}$$

one component of $\overrightarrow{AB} \times \overrightarrow{AD}$ correct

M1

all correct

A1cao

3

(ii) Area $ABCD = |\overrightarrow{AB} \times \overrightarrow{AD}| = \sqrt{8^2 + 14^2 + 10^2}$
FT their $|\overrightarrow{AB} \times \overrightarrow{AD}|$

M1

$$= \sqrt{64 + 196 + 100}$$

$$= \sqrt{360}$$

$$= \sqrt{36} \sqrt{10}$$

$$= 6\sqrt{10}$$

or $p = 6$

A1cso

2

(b) $\overrightarrow{AB} \times \overrightarrow{AD}$ is perpendicular to plane $ABCD$
used for $\mathbf{v}$ in vector line equation

M1

Hence direction ratios of line = $-8 : 14 : 10 = -4 : 7 : 5$

or any multiple of $\begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$

A1

M is mid-point of either diagonal = $\left(\frac{1+7}{2}, \frac{0+2}{2}, \frac{2+4}{2}\right)$
mid-point calculation

B1

$$= (4, 1, 3)$$

Hence line is $\left(\mathbf{r} - \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}\right) \times \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

All correct or equivalent multiple of $\begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$

A1cso

4

- (c) Perpendicular vector to $\Pi = \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix}$
 $\Rightarrow \Pi$ has equation $-4x + 7y + 5z = c$
ft their perpendicular vector

M1

Through $(6, 5, 17) \Rightarrow c = -4(6) + 7(5) + 5(17)$
using $(6, 5, 17)$

$$= -24 + 35 + 85 = 96$$

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A1

Equation is $-4x + 7y + 5z = 96$
ACF

$x = 4 - 4t; y = 1 + 7t; z = 3 + 5t$
parametric form of line

B1F

Line meets plane when
 $-4(4 - 4t) + 7(1 + 7t) + 5(3 + 5t) = 96$
 $-16 + 16t + 7 + 49t + 15 + 25t = 96$
 $90t = 90$

$$t = 1$$

substitution of parametric form and attempt to solve for t

M1

$\Rightarrow$ point of intersection = (0, 8, 8)
correct point of intersection

A1cao

6

Alternative 1:

$$\mathbf{r} = \begin{bmatrix} 6 \\ 5 \\ 17 \end{bmatrix} + s \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$$

$\mathbf{r} = \mathbf{a} + s\mathbf{d}_1 + t\mathbf{d}_2$ fully correct

(M1)

$$\begin{vmatrix} i & 2+2s+4t & -4 \\ j & 4-s+3t & 7 \\ k & 14+3s-t & 5 \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Substitute in $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = 0$

(m1)

Use of $\mathbf{r} - \mathbf{a}$ in parametric form – simplified

(B1F)

$$-26s + 22t = 78$$

$$22s + 16t = -66$$

$$10s + 40t = -30$$

Three correct equations obtained from vector product-terms collected

(A1)

$$s = -3, t = 0$$

Correctly solving equations to get both parameters

(M1)

$$\begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}$$

Correct point of intersection

(A1)

(6)

Alternative 2:

$$\mathbf{r} = \begin{bmatrix} 6 \\ 5 \\ 17 \end{bmatrix} + s \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$$

$\mathbf{r} = \mathbf{a} + s\mathbf{d}_1 + t\mathbf{d}_2$ fully correct

(M1)

$$\mathbf{r} = \begin{pmatrix} 4-4p \\ 1+7p \\ 3+5p \end{pmatrix}$$

Parametric form of line

(B1F)

$$2s + 4t + 4p = -2$$

$$-s + 3t - 7p = -4$$

$$3s - t - 5p = -14$$

Equating components, simplifying and attempting to solve-must at least reduce to 2 equations in two unknowns

(m1)

$$p = 2$$

Solving equations-one parameter correct

(M1)

$$t = 0 \text{ and } s = -3$$

All values correct

(A1)

$$\begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix}$$

Correct point of intersection

(A1)

(6)

[15]

Q2.

(a)

$$\left. \begin{array}{l} \text{direction ratios of line} = p : 3 : -1 \\ \text{normal to plane} = 1 : 1 : 2 \end{array} \right\} \begin{array}{l} \text{not} \\ \text{equal} \end{array}$$

Accept not parallel or showing vector product is non zero

B1

1

(b)
$$\begin{aligned} x &= 3 + pt \\ y &= q + 3t \\ z &= 1 - t \end{aligned}$$

Parametric form seen

M1

$$\text{Meets plane} \Rightarrow 3 + pt + q + 3t + 2(1 - t) = 10$$

$$\Rightarrow (5 + q) + t(p + 1) = 10$$

Correct substitution in plane

A1

$$\text{Within plane} \Rightarrow q = 5, p = -1$$

M1 – Finding one correct value

A1 – Both values correct

M1A1

4

Alternative:

Point $(3, q, 1)$ is common to line and plane

$$\text{Hence } \begin{pmatrix} 3 \\ q \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10 \text{ which gives } q = 5$$

Uses common point to find q

(M1A1)

Another point common to both is $(3 + p, 8, 0)$

$$\text{Hence } \begin{pmatrix} 3+p \\ 8 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10 \text{ which gives } p = -1$$

Use of second point and value of q to find p or

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consideration of scalar product

$$\begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

(M1A1)

(4)

$$(c) \quad (i) \quad \mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \quad \mathbf{d} = \begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix}$$

Let α be angle between normal and direction ratios
(plane) (line)

$$\mathbf{n} \cdot \mathbf{d} = p + 1$$

$\mathbf{n} \cdot \mathbf{d}$ correct

M1

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \cos \alpha = \frac{\pm 1}{\sqrt{6}}$$

Correct $\cos \alpha$ stated or implied

B1

$$\Rightarrow \frac{p+1}{\sqrt{6}\sqrt{p^2+10}} = \frac{\pm 1}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and** attempting to solve for p (condone missing $\pm$)
Dependent on first M1 – fully correct for A1

m1A1

$$\Rightarrow (p+1)^2 = p^2 + 10$$

$$\Rightarrow p^2 + 2p + 1 = p^2 + 10$$

$$\Rightarrow 2p = 9 \text{ giving } p = 4.5$$

CAO

A1

5

Alternative:

$$|n \times d| = \sqrt{49 + (1+2p)^2 + (3-p)^2}$$

$|n \times d|$ correct

(M1)

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \sin \alpha = \frac{\sqrt{5}}{\sqrt{6}}$$

Correct $\sin \alpha$ stated or implied. All rights reserved.

(B1)

$$\frac{\sqrt{49 + (1+2p)^2 + (3-p)^2}}{\sqrt{6}\sqrt{p^2+10}} = \frac{\sqrt{5}}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and** attempting to solve for p . Dependent on first M1 – fully correct for A1

(m1A1)

Leading to $p = 4.5$

CAO

(A1)

(5)

(ii) $z = 2 \Rightarrow t = -1 \Rightarrow x = -1.5$

$$p = 4.5 \quad y = q - 3$$

$$\Rightarrow -1.5 + q - 3 + 4 = 10$$

Attempt to form an equation for q using t = -1

M1

$$q = 10.5$$

CAO

A1

2

[12]

Q3.

(a) Use of scalar product on

$$\begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} \text{ and } \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}$$

M1

$$\text{Sc.Prod.} = \pm 21;$$

B1

$$\text{Moduli } \sqrt{54} \text{ and } \sqrt{26} \text{ correct}$$

Accept 7.348... & 5.099...

B1

$$\text{AWRT } 56^\circ$$

From correct working

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A1

(b) $2x + y + 7t = 10$ and $3x + y - 4t = 7$ noted or used

M1

Eliminating (say) y to get x as a fn. of t

M1

$$x = 11t - 3$$

CAO

A1

Subst^g. back for y

M1

$$y = 16 - 29t$$

CAO

A1

$$\frac{x+3}{11} = \frac{y-16}{-29} = z \quad (=t)$$

B1 ft

6

(c) Attempt at either $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} = 10$ or $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} = 7$

M1

Solving either

$$40 + 2\lambda - 1 + 9\lambda + 49 + 28\lambda = 10 \text{ or}$$

$$60 + 3\lambda - 1 + 9\lambda - 28 - 16\lambda = 7$$

M1

$$\lambda_1 = -2$$

$$\lambda_2 = 6$$

A1A1

$$P = (18, -19, -1) \text{ and } Q = (26, 53, 31)$$

$$PQ = \sqrt{8^2 + 72^2 + 32^2} = \sqrt{6272} = 56\sqrt{2}$$

NB P, Q not required:

$$d = |\lambda_1 - \lambda_2| \times |\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}| = 8 \times 7\sqrt{2} = 56\sqrt{2} \quad \text{M1 A1}$$

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6

[16]

Q4.

(a) $\Pi_1 : 6x + 2y + 9z = 5$

$$\Pi_2 : 10x - y - 11z = 4$$

Both

B1

1

(b) **Method 1**

$$\text{E.g. } \Pi_1 + 2\Pi_2 \Rightarrow 13(2x - z = 1)$$

M1

$$\frac{x}{1} = \frac{z+1}{2} = \lambda$$

Parametrisation attempt

M1

A1

Using 1st two to find 3rd in terms of λ :

$$x = \lambda, z = 2\lambda - 1 \Rightarrow y = 7 - 12\lambda$$

M1

$$\mathbf{r} = \begin{bmatrix} 0 \\ 7 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -12 \\ 2 \end{bmatrix}$$

Any correct vector eqn.form

A1

5

Method 2

$$(0, 7, -1) \text{ or } \left(\frac{7}{12}, 0, \frac{1}{6}\right) \text{ or } \left(\frac{1}{2}, 1, 0\right)$$

Finding a pt. on the line

(M1A1)

$$\begin{bmatrix} 6 \\ 2 \\ 9 \end{bmatrix} \times \begin{bmatrix} 10 \\ -1 \\ -11 \end{bmatrix} = (\pm)13 \begin{bmatrix} 1 \\ -12 \\ 2 \end{bmatrix}$$

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(M1)

Finding a d.v.

(A1)

$$\mathbf{r} = \begin{bmatrix} 0 \\ 7 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -12 \\ 2 \end{bmatrix}$$

Any correct vector eqn.form

(A1)

- (c) Substⁿ. $x = \lambda, y = 7 - 12\lambda, z = 2\lambda - 1$
into $5x + 3y + 11z = 28$

M1

Solving a linear eqn. in λ :

M1

$$\lambda = -2$$

CAO

A1

$$(-2, 31, -5)$$

FT their λ in their line eqn.

B1✓

4

(d) $\mathbf{n} = \begin{bmatrix} 1 \\ -12 \\ 2 \end{bmatrix}$

FT when used as part of a plane eqn, not a line

B1✓

$$\mathbf{d} = \begin{bmatrix} 4 \\ 1 \\ 9 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -12 \\ 2 \end{bmatrix} = 10$$

Attempted

M1

$$\mathbf{r} \cdot \begin{bmatrix} 1 \\ -12 \\ 2 \end{bmatrix} = 10$$

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Any correct vector eqn. form

FT

A1✓

3

$$\text{Alt. } \mathbf{r} = \begin{bmatrix} 4 \\ 1 \\ 9 \end{bmatrix} + \lambda \begin{bmatrix} 6 \\ 2 \\ 9 \end{bmatrix} + \begin{bmatrix} 10 \\ -1 \\ -11 \end{bmatrix}$$

Plane eqn. attempt

(M1)

Point + at least 1 d.v.

(A1)

All correct

(A1)

Q5.

- (a) Use of $\sin$ or $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$

using $\begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

M1

Numerator = 16, denominator = 21

Allow numerator $\sqrt{185}$

B1B1

$$\sin \theta = \frac{16}{21} \Rightarrow \theta \approx 49.6^\circ$$

Allow AWRT 49.6

A1

4

(b) $\begin{bmatrix} 2\lambda + 1 \\ \lambda + 2 \\ 2\lambda - 7 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix} = 37$

M1

$$6\lambda + 3 - 2\lambda - 4 + 12\lambda - 42 = 37$$

with attempt to solve

m1

$$\dots \Rightarrow \lambda = 5$$

A1

giving $P = (11, 7, 3)$

ft wrong value of λ

B1F

4

(c) (i) Use of the vectors $\begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

M1

Vector product attempted

OE

m1

Required vector is $\begin{bmatrix} -10 \\ 6 \\ 7 \end{bmatrix}$
Or a non-zero multiple

A1

3

(ii) $a = \begin{bmatrix} 11 \\ 7 \\ 3 \end{bmatrix}$

ft wrong answer in (b)

B1F

$b = \begin{bmatrix} -10 \\ 6 \\ 7 \end{bmatrix} \times \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 50 \\ 81 \\ 2 \end{bmatrix}$

Or a non-zero multiple; ft wrong answer to (c)(i)

M1A1F

Fully correct equation for L'

A1

4

[15]

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Q6.

(a) (i) $\begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix} = 0$

Equating dot product to zero

M1

$24p + 6 - 3p + 6 + 2 = 0$

Solving a linear eqn. in t

M1

$p = -\frac{2}{3}$
CAO

A1

3

$$(ii) \begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} = m \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix}$$

M1

$$m = 2, p = \frac{1}{2}$$

A1,A1

3

$$\text{ALT} \begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} \times \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix} = \mathbf{0} \quad \begin{bmatrix} 1-2p \\ 8p-4 \\ 18p-9 \end{bmatrix}$$

(M1A1)

$$p = \frac{1}{2}$$

(A1)

$$(b) (i) 6x - 3y + 2z = 42, 17x + 2y + z = -7$$

B1,B1

2

$$(ii) \text{ eg } 2. \textcircled{2} - \textcircled{1} : 7 [4x + y = -8]$$

Eliminating one variable; correct

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M1 A1

$$\frac{x+2}{-1} = \frac{y}{4} = \lambda$$

Parametrisation attempt

M1

Substituting back to find 3rd variable

m1

$$z = 9\lambda + 27$$

A1

$$\mathbf{r} = \begin{bmatrix} 0 \\ -8 \\ 9 \end{bmatrix}, \begin{bmatrix} 1 \\ -12 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} -2 \\ 0 \\ 27 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix}$$

ft (any pt. on l)

B1

OR method for finding the d. v. $\begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix}$

(M2A1)

Method for finding any pt. on l

(M1A1)

Putting them together as a line eqn.

ft

(B1)

6

- (c) $\mathbf{r} = \mathbf{a} + u \mathbf{d}_1 + v \mathbf{d}_2$
 where $\mathbf{a}$ is an pt. on plane
ft their previous a; or (30, 7, 30)

$\mathbf{d}_1 =$ any d. v.
ft their previous d. v.

$\mathbf{d}_2 = \begin{bmatrix} 30 \\ 7 \\ 30 \end{bmatrix}$ any $\mathbf{a}$ in plane
ft if is clear where it has come from

2

[16]

Q7.

- (a) Clearly identifying $\mathbf{n} = \begin{bmatrix} 9 \\ -8 \\ 72 \end{bmatrix}$

B1

$$\mathbf{d} = \begin{bmatrix} 9 \\ -8 \\ 72 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 10 \\ 1 \end{bmatrix} = 10$$

M1A1

- (b) Use of $\frac{\text{Sc. prod. of normals}}{\text{prod. of their moduli}}$
 Must be $(9i - 8j + 72k)$, $(i + j + k)$
 or their n from (a)

M1

$$N^r = 73$$

$$D^{rr} = 73\sqrt{3} \text{ or } \sqrt{15987}$$

Fit their n from (a) only

B1ftB1ft

$$\cos \theta = \frac{1}{\sqrt{3}}$$

CAO Allow unsimplified exact forms

A1

4

[7]

Q8.

(a) Either $\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} \times \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = -3 \begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix}$

Correct vector product only here

M1A1

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Explaining that d.v. of L is in dirn. of Π 's nml. $\Rightarrow L \perp \Pi$

B1

Or $\begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} = 0$ and $\begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = 0$

(M1)(A1)

Explaining that d.v. of L is $\perp$ to 2 (non-//) vectors in $\Pi \Rightarrow L \perp \Pi$
 Not just stating

(B1)

3

(b) E.g. $6 \times \textcircled{1} - \textcircled{2}$: $46 = 34p + 27q$

Eliminating r from any pair of eqns.

M1

$$2 \times \textcircled{1} - \textcircled{3}: -57 = 21p + 6q$$

$$\textcircled{2} - 3 \times \textcircled{3}: -217 = 29p - 9q$$

Any 2 correct eqns (1 mark each)

A1A1

$$2 \times \textcircled{4} + 9 \times \textcircled{5}: 605 = -121p$$

Solving a 2×2 system (any means) in order to get values for p, q, r

M1

$$p = -5, q = 8, r = -1$$

All 3 ✓ CAO

A1

5

(c) $7 + 6t = -2 + 5\lambda + \mu$
(i) $8 + 2t = 0 + 3\lambda + 6\mu$
 $50 - 9t = -25 + 4\lambda + 2\mu$

Including re-arrangement attempt

M1

$$\begin{aligned} 9 &= -6t + 5\lambda + \mu \\ \rightarrow 8 &= -2t + 3\lambda + 6\mu \\ 75 &= 9t + 4\lambda + 2\mu \end{aligned}$$

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i.e. the above system with
 $p = -t, q = \lambda$ and $r = \mu$

A1

2

(ii) Substg. $t = 5$ into L 's eqn.

Or $\lambda = 8$ **and** $\mu = -1$ into IT 's eqn.

M1

$$P = (37, 18, 5)$$

CAO

A1

2

[12]

Q9.

(a) (i) $\mathbf{n} = (3\mathbf{j} + \mathbf{k}) \times (4\mathbf{i} - \mathbf{j}) = \mathbf{i} + 4\mathbf{j} - 12\mathbf{k}$

M1A1

2

Alternative:

$$\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \quad \text{and/or} \quad \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

(M1)

$$\Rightarrow c = -3b, b = 4a \text{ i.e. } a(\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$$

(A1)

(2)

(ii) $d = (2\mathbf{i} + 5\mathbf{j} + \mathbf{k}) \cdot (\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}) = 10$

Their $\mathbf{n}$

M1A1 ✓

2

(b) Either substituting.

$$\mathbf{r} = \begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix} \text{ into } \left(\mathbf{r} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$$

M1

And showing result is 0

A1

2

Or

Explaining p.v. and d.v. of line

Give the M1 for either

M1A1

2

(c) Subst^g. $\begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix}$ into their plane eqn.

M1

Solving a linear eqn. in t

$$2 + 3t + 8 - 60 + 12t = 10$$

dM1

$$t = 4$$

A1 ✓

Pt. of intersection at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
CAO

B1

4

First alternative (c)

Uses (b)'s given form for L and the original form for Π

Setting
$$\begin{bmatrix} 2+3t \\ 2 \\ 5-t \end{bmatrix} = \begin{bmatrix} 2+4\mu \\ 5+3\lambda-\mu \\ 1+\lambda \end{bmatrix}$$

Must make an attempt to solve

(M1)

$$\mu = \frac{3}{4}t, \mu = 3+3\lambda, \lambda = 4-t$$

(A1)

Eliminating λ and μ (e.g.) $\Rightarrow t = 4$

(dM1)

Pt. of i/sctn. at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
CAO

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4

Second alternative (c)

Uses other given form for L

$$\left(\begin{bmatrix} 2+4\mu \\ 5+3\lambda-\mu \\ 1+\lambda \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} \text{ gives}$$

$$\begin{bmatrix} 3+4\mu \\ 3+3\lambda-\mu \\ \lambda-5 \end{bmatrix} \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -3-3\lambda+\mu \\ 3\lambda+4\mu-12 \\ 3\mu-9\lambda-9 \end{bmatrix}$$

(M1 A1)

Equating to $\mathbf{0}$ and solving

$$\Rightarrow \lambda = 0, \mu = 3$$

(M1)

Pt. of intersection at $14\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
CAO

(A1)

4

[10]



EXAM PAPERS PRACTICE

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Examiner reports

Q1.

Generally there was good response, with all students able to score marks in the earlier part of the question. The vector product evaluation through a determinant is well known; nevertheless, a handful of students made sign errors. Students also know how to apply the determinant to finding the area of a parallelogram. Many students did not appreciate the link between the parts of this question. Thus the determinant evaluated in part (a) was re-evaluated in part (b) albeit with the diagonals of the parallelogram instead. With regard to vector notation, students should realise that they must make key vectors clear, eg the

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

zero vector written as $\mathbf{0}$ or $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Many students did not find the coordinates of $\mathbf{M}$ by using the mid-point result from Core 1 but instead chose to find the intersection point of two lines. Part (c) proved most challenging. Those who used the scalar product or Cartesian form of the plane achieved the greatest success in finding the point of intersection of the line and plane.

Q2.

This topic of equations of lines and planes is not well understood by students. Surprisingly, part (a) was not always answered correctly with arguments concerning the scalar product often wrongly used. Part (b) was most often successful when two points on the line were used or if the scalar product of the normal to the plane and direction vector of the line was set equal to zero and solved to find p . Part (c)(i) proved very discriminating with students using both scalar and vector products in tandem but not using the correct trigonometric ratio and confusing sine and cosine.

Q3.

This was another high-scoring question for many candidates, especially since the three parts were essentially completely independent of each other's results. Part (a) was straightforward and routine, though rather a lot of candidates lost marks by failing to give the acute angle required, subtracting the correct one from 90° , mixing up sines and cosines in the working, missing the minus sign in front of the "4" in the second of the two given normal vectors, or making numerical slips in adding up the three squares in each of the moduli.

Q4.

Around a third of the candidature attempted little beyond part (a), which was very surprising.

Of the majority who did proceed well into the question, part(b) was usually done well, although there were many sign errors that arose in the vector product of the two normals. Another surprise was the lack of appreciation that there was a factor of 13 which could be ignored here.

Of those who realised in part(c) that they simply had to substitute part(b)'s answer into Π_3 's equation, a slight majority had incorrect components to work with, and this made follow-through marking difficult.

In part(d), only a very few spotted that they could answer this without any need to rely on

previous answers, as a plane equation in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}_1 + \mu \mathbf{d}_2$ could be written straight down using the two normals given in the question. The most popular form was, of course, $\mathbf{r} \cdot \mathbf{n} = d$, although a lot of candidates opted for a line equation here instead.

Q5.

The first two parts of this question gave almost all candidates the chance to use techniques with which they were clearly very familiar. The last mark in part (a) was often lost by a failure to subtract the inverse cosine from 90° , or equivalently to use the inverse sine.

In part (c)(i), many candidates saw that the required vector could be found by using a vector product, but others formed linear equations which they then struggled to solve. Part (c)(ii) was beyond the grasp of most candidates, though they could pick up one mark by forming an equation of a line passing through P . Very few candidates saw the need to form another vector product from the direction vector of the line L and the vector found in part (c)(i).

Q6.

The structure of this question proved very helpful to candidates, and those who were reasonably careful could generally score at least 13 of this question's 16 marks without any difficulty. The main hurdle arose where many candidates failed to realise in part (b) that they had already worked out the eigenvalues of $\mathbf{M}$ in part (a), and thus they went ahead and started all over again in an effort to find, and then solve, a cubic characteristic equation. This was all a bit of a waste of time and frequently not as successful as their former effort.

Q7.

A similar fault was all too common here in part (a), even amongst those who presumably knew what they were doing. A lot of candidates correctly found $d = 10$ from the use of the vector $9\mathbf{i} - 8\mathbf{j} + 72\mathbf{k}$, yet still lost a mark by nowhere identifying this as the " $\mathbf{n}$ " of the question ... not even to the extent of writing down the answer in the form requested. This is just carelessness. Apart from a small minority who failed to identify $\mathbf{n}$ correctly, part (b) was found to be very straightforward indeed.

Q8.

There were several easy marks to be had in this question, and the rest was helpfully sign-posted. Part (a) was routine ... or it would have been had not, again, some sort of explanation been required for full marks. Approaches were evenly split between using the scalar-product and the vector-product. In the former case, a lot of candidates seemed to think they only needed to show that the line's direction vector was perpendicular to **one** vector parallel to the plane. In the second case, many candidates were content merely to find the vector product of the two vectors "in" (parallel to) the plane; others noted that it was a multiple of the d.v. of L without further comment. For the 3rd mark on this part of the question, we wanted an explanation as to *why* their scalar- or vector-product work established the given outcome – it is not enough for candidates merely to repeat the result that we gave them. *Given* results always require more of candidates by way of 'dotting the i's and crossing the t's'.

The more discerning teachers and students of MFP4 will understand why part (b) was set up as it was, since this is the only way to force candidates not just to resort to calculator-output generated answers. In part (c), remarkably few got past equating the

equations of L and Π ; the majority of candidates had no real idea what to do and so there were lots of answers seen in which the point of intersection was taken to be $(-5, 8, -1)$.

Q9.

This was one of the questions which candidates with a weak grasp of the underlying ideas behind topics found difficult. Part (a)(i) can be done either by a single vector product or by two scalar products. A large number of candidates then used one of the plane's two direction vectors, rather than the position vector of the point $(2, 5, 1)$ which lies in the plane Π , to find the scalar constant d required. This is one of several fundamental errors made by candidates.

Part (b) was a straightforward test of the alternate form of a line equation, yet it was frequently answered in unconvincing ways. The instruction to "verify" – that is, to substitute the given expression for $\mathbf{r}$ into the given equation for L – was clearly not understood by many. The perfectly allowable alternative approach is to explain the role of the line's direction vector and show that the point $(2, 2, 5)$ is a point on the line. Many candidates managed one or the other but there was often a confusing amount of vector algebra and arithmetic presented with no apparent purpose and absolutely no explanation. Such vague approaches usually gained one of the two marks, but not both.

Part (c) was generally done quite well, as there are at least three approaches, at least one of which does not require successful progress in any previous part of the question.